CIFAR-10 Residual Networks: A Finite Crop and Mixup Study

Deep Learning Project 1 - Fall 2026
Author and Kaggle team: Zihan Xu

Public code, weights and executed notebook:
https://github.com/Zihan-Xu/DL-project1

1. Overview and findings

This experiment investigates training augmentation while retaining a previously tested residual architecture under the five-million-parameter limit. Two new recipes - random crop and random crop plus Mixup - each completed 80 epochs at seeds 142,143,144. Three previously accepted, matched-budget runs supply the no-crop control. All six new runs passed independent CPU replay before selection.

Measure	Result and interpretation
Selected new recipe	crop_mixup / tta; three-seed validation mean 95.7800%; seed SD 0.0872 percentage points
Parameter count	2,436,346 trainable parameters, including the classifier
Dual residual candidate	96.3333% three-seed validation mean; 4,872,692 total parameters across both branches. New official score unavailable.
Final new candidate	From scratch on all 50,000 course training images; fixed seed 42; last of80 epochs
New final test accuracy	Unavailable: the new CSV has not been officially scored; no independent local holdout remains after the full50k fit
Previous Kaggle result	0.87900 (87.90%), user reported for long_111/full50k/TTA. Remote submitted bytes were not independently retrieved.

The new recipe's validation mean does not exceed the earlier 240-epoch recipe's96.04%. This comparison uses different training budgets and does not establish a Kaggle improvement. The controlled80-epoch comparison appears on page2. Local validation and official hidden-test scores remain separate.

2. Architecture and rationale

The width3 CIFAR ResNet-20 has channels 48/96/192 and three two-convolution residual blocks per stage. Each block combines its convolution-BN-ReLU residual branch with a skip connection; transitions use learned1x1 projections. Fixed3x3 binomial BlurPool filters precede stride-two sampling. Global average pooling feeds a ten-class linear head. Residual connections support optimization [1], while low-pass filtering reduces sensitivity to sampling alignment [2].

Widening provides capacity within the parameter budget, at greater compute cost. BlurPool adds filtering without trainable parameters. This round holds architecture fixed to isolate augmentation changes; it uses no rotation, dropout, stochastic depth, attention, teacher, pretrained weights or external dataset. Raw inference is approximately436.1 million MACs; two-view inference doubles the forward-pass work.

The 0.87900 reference uses the same widened BlurPool backbone with rotation (+/-10 degrees, p0.5), dropout 0.1 [11] and per-sample stochastic depth up to 0.1 [12], trained for 240 epochs. Its weights and full recipe remain in the previous long111_full50k_v2 release. The reference score does not belong to the new crop/Mixup model.

3. Matched-budget protocol and results

Research training uses40,000 images and the same stratified5,000-image validation set as earlier studies. Normalization is computed on those40k training images only. The accepted confirm_000 controls use the same seeds, optimizer, schedule, sampler and80-epoch budget. Each training epoch covers every training image once. Checkpoint selection takes the highest raw-validation correct count, with earliest-epoch tie breaking; it does not stop training early.

Both new recipes add reflection padding 4 followed by a uniform32x32 integer crop, then the original horizontal flip(p0.5) and Cutout 12(p0.5). Crop+Mixup applies a Beta(0.2,0.2) mixture with probability 0.5 per batch, using identical mixture weights for images and cross-entropy labels [3,4]. Crop, Mixup, sampling and existing augmentations have separate deterministic random streams.

Recipe	Raw mean %	Flip mean %	Raw / flip seed SD (pp)
control	94.8267	95.2333	0.1553 / 0.1332
crop	95.2200	95.6600	0.1114 / 0.1833
crop_mixup	95.4667	95.7800	0.1665 / 0.0872

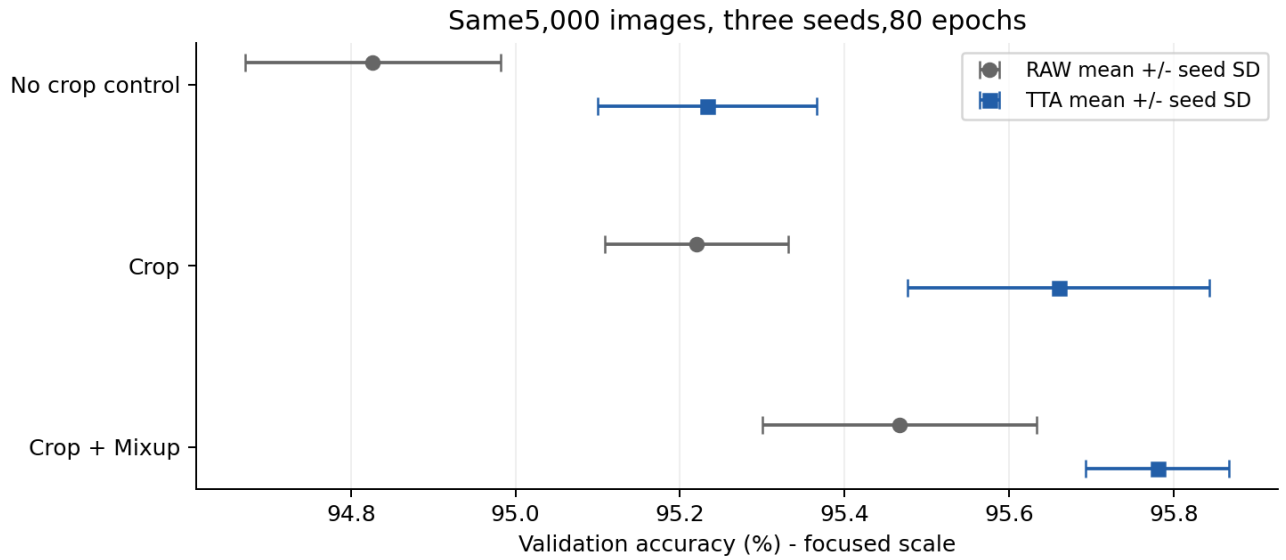


Figure 1. Complete three-seed means on the same 5,000-image validation set. Error bars are seed SD, not confidence intervals. The accuracy axis is focused; exact values appear above.

Raw and horizontal-flip probability averaging use the same fixed checkpoint for each run. Selection ranks the complete three-seed mean over the four new configuration/policy choices; ties favor raw, then no Mixup. Deployment seed42 was fixed before results. The experiment does not select the most favorable individual seed.

4. Evidence strength and completed full-data fit

Comparison / inference	Mean delta (pp)	Paired95% t interval	Holm p
crop - control / raw	+0.3933	[-0.1994, 0.9861]	0.3117
crop - control / tta	+0.4267	[-0.3531, 1.2064]	0.3117
crop_mixup - control / raw	+0.6400	[-0.1345, 1.4145]	0.2832
crop_mixup - control / tta	+0.5467	[0.2159, 0.8775]	0.1153
crop_mixup - crop / raw	+0.2467	[0.0459, 0.4475]	0.1699
crop_mixup - crop / tta	+0.1200	[-0.3987, 0.6387]	0.4244

The intervals quantify variation across three paired training seeds, not independent data replication. Six exploratory comparisons use Holm adjustment. The same validation set has been reused for architecture and checkpoint selection, so both ranking and significance should be treated cautiously. More data-independent confirmation would be needed for a strong generalization claim.

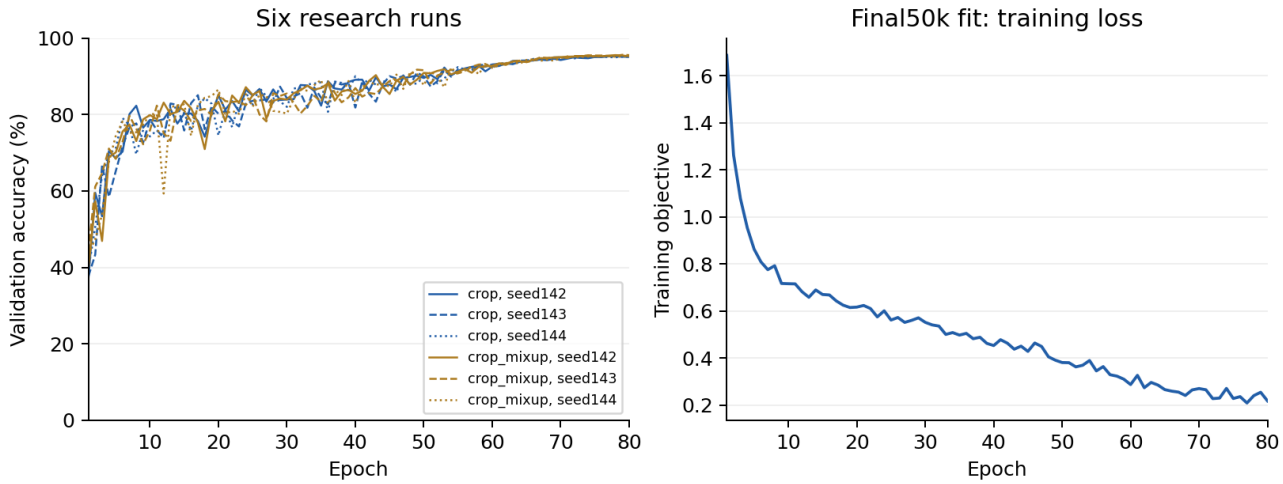


Figure 2. All six research training trajectories and the final 50k training loss. Research axes show the full0-100% range. Mixup training accuracy is not compared because labels are mixed.

After acceptance, crop_mixup was trained from scratch on all 50,000 images for80 epochs using fixed seed 42. SGD uses initial LR 0.1, momentum 0.9, weight decay 0.0005, batch 128 and a cosine schedule without warmup. Every epoch has 391 updates, including a final80-image batch:31,280 updates and4,000,000 image presentations overall. The last prescribed epoch supplies final weights; no former holdout is used for early stopping.

RGB mean/std are recomputed from all 50k training images. Training uses BF16, inference uses FP32, TF32 is disabled and deterministic algorithms are enabled. The recorded environment is Python3.13.5, PyTorch2.11.0+cu128, NumPy2.1.3 and RTX5070Ti. Full configuration, environment and exact source hashes are included.

5. Separately registered dual-residual experiment

Before evaluating any combination, a secondary plan fixed equal probability averaging between the complete primary-study winner and the old long_111/TTA branch. It uses seeds 142, 143 and 144 paired within each combination. No mixture weight, temperature or alternate primary recipe is searched. Recommendation requires all three independent CPU replays and a mean higher than both individual recipes. This secondary study follows primary results and uses the same validation set; it is not a new holdout [13].

Seed	Old correct	New correct	Dual correct	Validation N
142	4803	4787	4818	5000
143	4794	4786	4817	5000
144	4809	4794	4815	5000

The complete dual mean is **96.3333%**, with seed SD 0.0306 percentage points. The paired mean change over the old240-epoch recipe is +0.2933 points, with a95% seed-variation t interval [-0.1292, 0.7159]. The fixed all-seed recommendation criterion passed: True. The three replays reproduce saved combined probabilities and labels independently on CPU.

The deployable dual model contains both complete residual networks and both classifier heads. All4,872,692 parameters count toward the5M limit, including parameters that are not updated during inference. Each branch is trained from scratch on course data; there are no external pretrained weights. The final new branch costs80 epochs and the reused reference costs240 epochs, both on50k images with seed42. Thus the combined training cost is80+240 epochs, not80. Each branch normalizes inputs with its recorded training statistics; probabilities are averaged0.5/0.5 after the fixed branch-specific inference policies.

The reviewed architecture rule permits custom residual CNN components and requires all trainable parameters to fit under5M; it contains no explicit ensemble prohibition. The dual construction follows this reading, without special staff approval. New weights and predictions are kept local until the competition ends, respecting the ban on sharing between teams during the competition.

6. Submission, reproducibility and lessons

Four distinct, validated CSV choices are supplied: the selected new policy, the other raw/flip policy using identical new weights, the fixed dual model, and the previous long_111 submission associated with the reported0.87900. Every file has exactly 2,000 unique sample-ordered IDs, columns id,label and integer labels 0-9. Metadata binds each prediction to model/config/policy and SHA-256. No Kaggle or Brightspace submission is made by this work.

The release includes final weights; six research checkpoints; architecture and augmentation implementations; runnable training, independent labeled-data evaluation and prediction; exact configuration/environment; complete epoch logs and saved validation predictions; comparison evidence; an executed review notebook; and the experiment spreadsheet. The portable engine retains the executed computational loop. Pixel-level augmentation, initialization, loss and gradient parity are checked; final prediction is independently replayed on CPU from the original test PNGs. No raw course images, credentials, messages or caches are published.

The principal lesson is to compare augmentation at equal compute and across all registered seeds. Training-time crop and label mixing may help generalization, but their empirical value depends on the completed comparison. Longer training, a new architecture and more augmentation simultaneously would obscure attribution and exceed the deadline budget. CutMix [5], AugMix [6], RandAugment [7] and WideResNet [8] remain researched but unexecuted alternatives in this round.

The remaining gap is official hidden-test evaluation. The earlier0.87900 and local validation scores refer to different populations; their difference is not enough to diagnose a particular distribution shift. The final 50k model has no unused local holdout, so a local accuracy must not be presented as its test score. An improved official score cannot be claimed until the new CSV is actually evaluated.

References

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